GROUP 5

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MDA3083 PREDICTIVE ANALYSIS

TOPIC: DATA REDUCTION AND BIG DATA

PRINCIPAL COMPONENT ANALYSIS (PCA) ON CUSTOMER SHOPPING BEHAVIOUR



Introduction

- **Big Data Challenges** → Large, complex consumer datasets create issues like dimensionality, multicollinearity, and inefficiency
- **Project Purpose** → Coursework for Predictive Analytics, focusing on Data Reduction and Big Data
- **Dataset** → Kaggle dataset with 3,900 observations and 16 variables (demographic + behavioural).
- **Method** → Principal Component Analysis (**PCA**) applied to reduce dimensionality and capture maximum variance
- **Tools** → Analysis conducted in Python using scikit-learn, pandas, matplotlib, and seaborn





Objectives

1. **Data Preprocessing** - Clean dataset, remove irrelevant features, encode categorical variables

2. **Principal Component Analysis** - Apply PCA to capture about 79.5% of the variance while studying the interaction between demographic and purchasing attributes

3. **Optimized Structures** - Build low-dimensional data for clustering, satisfaction modeling, and marketing applications



Data Description

Dataset Summary

Item	Description
Dataset Name	Shopping Behaviour Dataset
Number of Observations	3900
Number of Variables	16
Data Type	Numerical and Categorical
Domain	Retail Business
Analysis Method	Principal Component Analysis (PCA)

- **Observations** → 3,900 records
- **Variables** → 16 attributes (numerical + categorical)
- **Domain** → Retail business, customer shopping behaviour
- **Method** → PCA chosen for dimensionality reduction

- **Demographics** → Age, Gender.
- **Purchasing** → Item Purchased, Category, Purchase Amount.
- **Behaviour** → Review Rating, Previous Purchases, Subscription Status, Discount Applied.
- **Transaction** → Payment Method, Size, Color, Season.

Variable Description

Variable	Description	Type
Customer ID	Unique customer identifier	Integer
Age	Customer age	Numerical
Gender	Customer gender	Categorical
Item Purchased	Purchased product	Categorical
Category	Product category	Categorical
Purchase Amount (USD)	Total spending	Numerical
Location	Customer location	Categorical
Size	Product size	Categorical
Color	Product color	Categorical
Season	Season of purchase	Categorical
Review Rating	Customer rating	Numerical
Subscription Status	Subscription status	Categorical
Discount Applied	Discount applied	Categorical
Previous Purchases	Number of previous purchases	Numerical
Payment Method	Payment method	Categorical
Frequency of Purchases	Purchase frequency	Categorical

Data Description

DATA OVERVIEW

- **Dataset** → Shopping Behaviour Dataset from Kaggle.
- **Size** → 3,900 observations, 16 variables.
- **Type** → Mix of numerical and categorical attributes.
- **Domain** → Retail business, focusing on customer demographics and purchasing behaviour.
- **Purpose** → Suitable for PCA to identify key shopping behaviour factors.

DATA SOURCE

- **Origin** → Kaggle repository.
- **Content** → Customer demographics + purchasing information.
- **Purpose** → Provides foundation for analyzing shopping behaviour and applying PCA.

DATA SUITABILITY

- **Large Sample** → 3,900 observations ensure reliable analysis.
- **Mixed Variables** → Combination of numerical and categorical attributes.
- **Fit for PCA** → Suitable for dimensionality reduction to identify key shopping behaviour factors.

Methodology

1. **Data Cleaning & Feature Selection:**
Excluded non-behavioral variables (*Customer ID, Location, and Frequency of Purchase*)

2. **Categorical Data Transformation:**
Applied **Ordinal *Encoder*** to convert categorical variables into numerical format required for matrix calculations.

3. **Feature Standardization:**
Utilized ***StandardScaler*** to uniform the scale of all features to prevent large-magnitude variables from dominating.

Result & Discussion

Results Outcome: Successfully streamlined the data structure from 16 to 13 core behavioral features with verified zero missing values.

Results Outcome: Employed an **Ordinal *Encoder*** to turn categorical variables(e.g., *Gender, Category, Season*) into compact numerical indices (0,1,2) without inflating data dimensionality.

Results Outcome: Adjusted raw distributions such as *Age* (mean: 44.07) and *Purchase Amount* (mean: $59.76) to a mean of 0 and standard deviation of 1, ensuring fair variance distribution in PCA.

Methodology

Result & Discussion

4

Dimensionality Reduction (PCA Execution): Conducted eigenvalue decomposition on the standardized covariance matrix **($\Sigma = V\Lambda V^T$)**.

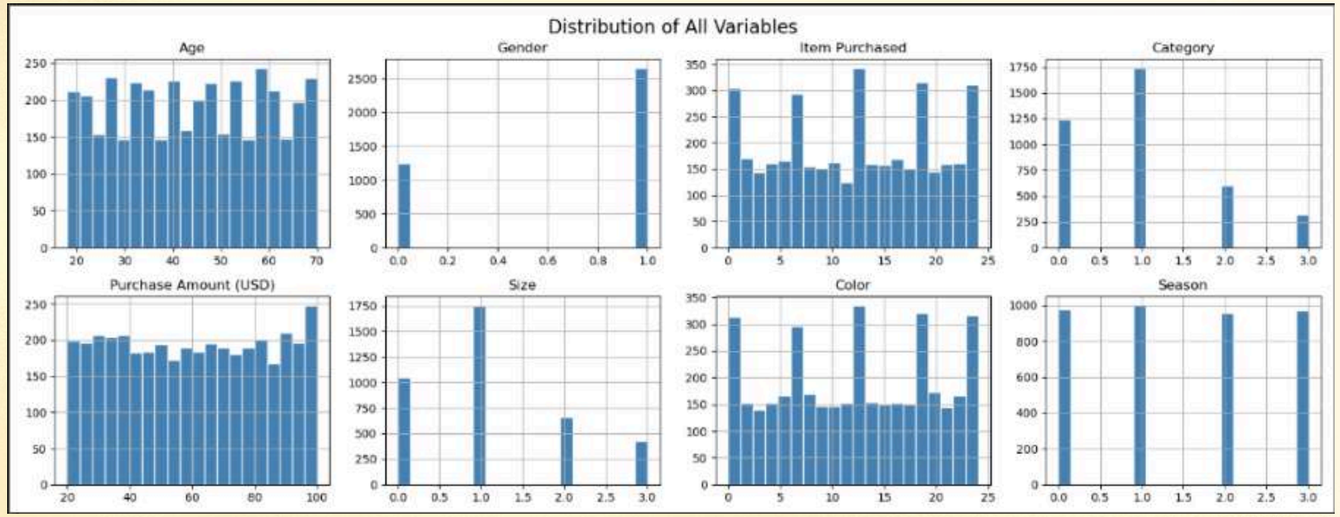
Results Outcome: Extracted **9 optimal Principal Components (PC1-PC9)** via Scree Plot analysis, reducing structural complexity by **43.75%** while capturing **79.52%** of total variance.

--- Descriptive Statistics Summary ---

	Age	Gender	Item Purchased	Category	Purchase Amount (USD)
count	3900.00	3900.00	3900.00	3900.0	3900.00
mean	44.07	0.68	12.04	1.0	59.76
std	15.21	0.47	7.20	0.9	23.69
min	18.00	0.00	0.00	0.0	20.00
25%	31.00	0.00	6.00	0.0	39.00
50%	44.00	1.00	12.00	1.0	60.00
75%	57.00	1.00	18.00	1.0	81.00
max	70.00	1.00	24.00	3.0	100.00

	Size	Color	Season	Review Rating	Subscription Status
count	3900.00	3900.00	3900.00	3900.00	3900.00
mean	1.12	12.11	1.49	3.75	0.27
std	0.93	7.22	1.12	0.72	0.44
min	0.00	0.00	0.00	2.50	0.00
25%	0.00	6.00	0.75	3.10	0.00
50%	1.00	12.00	1.00	3.70	0.00
75%	2.00	18.00	2.00	4.40	1.00
max	3.00	24.00	3.00	5.00	1.00

	Discount Applied	Previous Purchases	Payment Method
count	3900.00	3900.00	3900.00
mean	0.43	25.35	2.51
std	0.50	14.45	1.69
min	0.00	1.00	0.00
25%	0.00	13.00	1.00
50%	0.00	25.00	2.00
75%	1.00	38.00	4.00
max	1.00	50.00	5.00

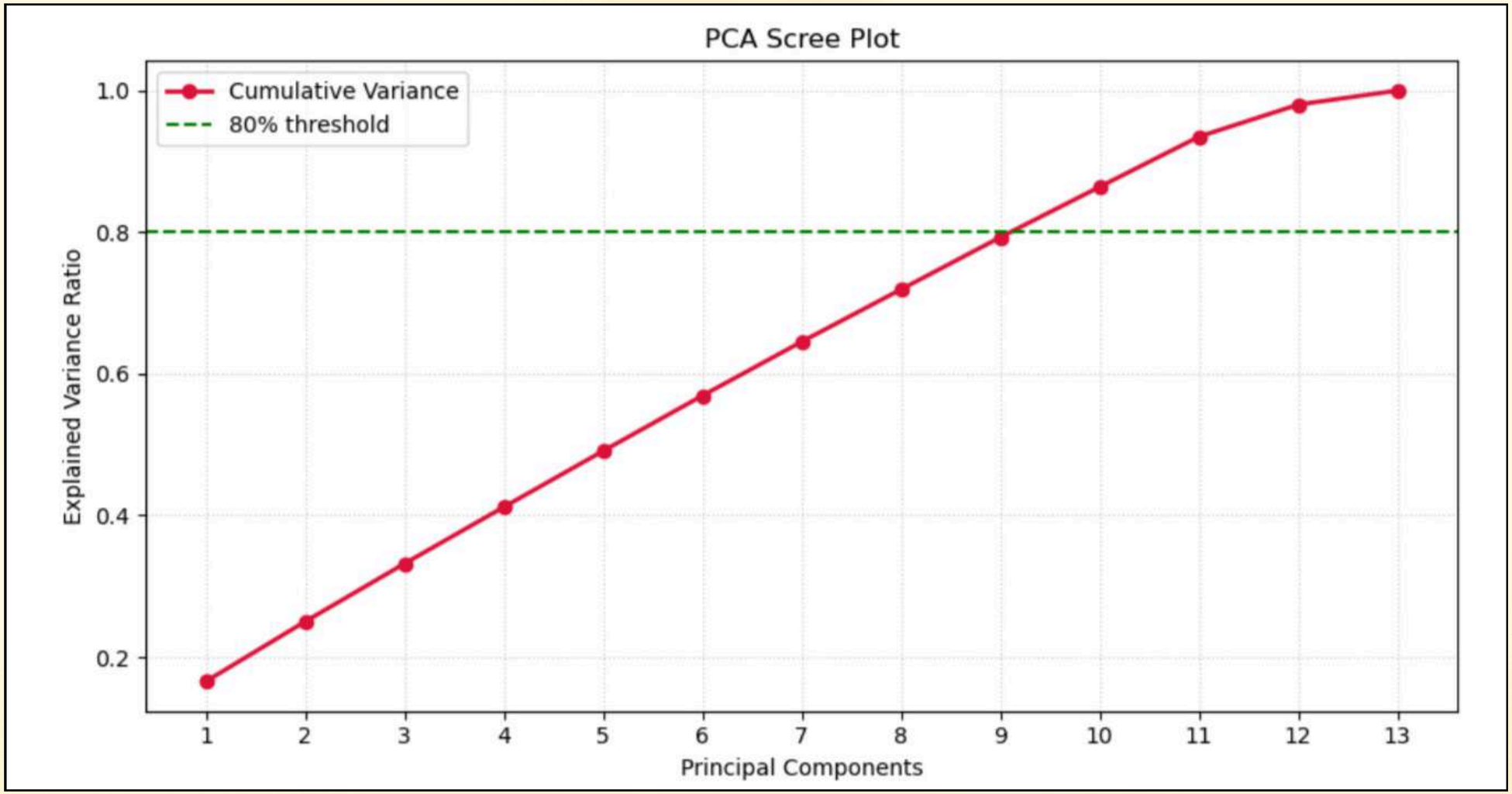




PCA Analysis







PCA Analysis

Component	Variance Explained (%)	Cumulative Variance (%)
PC1	16.57	16.57
PC2	8.46	25.03
PC3	8.18	33.21
PC4	7.99	41.20
PC5	7.92	49.12
PC6	7.81	56.93
PC7	7.62	64.55
PC8	7.39	71.94
PC9	7.36	79.52
PC10-13	13.63	100.00

Core Purchasing & Loyalty

- **PC 1 : Customer Incentive Profile (16.57%)** – Driven by discounts (0.62) and subscription status (0.57); captures promo responsive members.
- **PC 2: Magnitude & Sizing Trade-off (8.46%)** – Fit size (0.53) and spending (0.39) They spend a lot more money when they buy heavy items like winter coats or boots.
- **PC 3: History Contradiction (8.18%)** – Highlights a gap where new item selections (0.46) oppose older age (-0.41) .They are brand new, young customers buying items from the store for the first time.
- **PC 4: Brand Engagement Profile (7.99%)** – Links high numbers of previous purchases (0.53) with top review ratings (0.53) to highlight highly satisfied returning loyalists

Timing & Methods

- **PC 5: Seasonal Demographics (7.92%)** – Strongly maps seasonal changes (0.67) against younger customer age groups (-0.53) to track trend buyers.
- **PC 6: Transaction Preferences (7.81%)** – Dominated by preferred checkout payment methods (0.60) shifting dynamically across retail seasons (-0.51).
- **PC 7: Appearance vs. Spending (7.62%)** – Heavy focus on product color aesthetics (0.65) sharply contrasts against lower financial basket spend (-0.56).

Aesthetic & Assortment Preferences

- **PC 8: Size vs. Budget Limits (7.39%)** – Captures a consumer segment where high satisfaction with fit/size (0.59) occurs alongside restricted spending budgets (-0.57).
- **PC 9: Product Type vs. Evaluation (7.36%)** – Pairs specific individual item choices (0.50) directly with lower customer review ratings (-0.50)

Conclusion

- PCA successfully reduced the dataset from 13 variables to 9 principal components.
- The retained components explain **79.52%** of the total variance.
- The dimensionality reduction simplified the dataset while preserving important information.
- The reduced dataset can be used for customer segmentation, clustering, predictive modelling and targeted marketing strategies.



Poster Article Review

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Thank You

